1. The RV X has the following PMF

$$P(X=1)=0.3, P(X=2)=0.2, P(X=3)=0.4, P(X=4)=0.1$$

Calculate $E(X)$ and $E(2X^2+X)$

2. Let X be the number of heads in four tosses of a fair coin. Find E(X) and Var(X).

3. A fair coin is tossed repeatedly until a tail comes up. If X tosses are needed, then the Casino pays the gambler Y=X+1 dollars. What is the expected payout E(Y)?

4. An urn contains 9 $1 bills and 1 $100 bill. Let the random variable X be the total amount that results when two bills are drawn from the urn without replacement.

(a) Find the PMF.

(b) Find E(X) and Var(X).

5. The pdf of the RV X is given as follows. Find E(X) and Var(X)

$$f_X(x)=\begin{cases}10/x^2, & x>10\\ 0, & x\le 10\end{cases}$$

6. Cooper & McGillem Book

2-4.1 (a) (c)